RESEARCH PAPER

Solar energy harvesting on building's rooftops: A case of a Nigeria cosmopolitan city

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In this paper, the possibility of meeting the growing electrical energy demands of the city of Ibadan (one of the Nigeria cosmopolitan city) using the rooftop solar PV systems is investigated with the aim of determining the available rooftop for possible PV deployment. First, the optimal tilt angle as well as solar irradiation on tilted surfaces are determined for the city, Thereafter, the available roof area for possible solar PV- rooftops placement are analysed using reduction factor technique with the aid of ArcGIS, Google earth Imagery and population sampling techniques. Some of the key results reveal that the total roof area for the city of Ibadan is about 49.54 sq km while the available roof area for rooftop PV deployment is about 7.54 sq km with a maximum installation capacity of 1734.8 MWp, at an optimum fixed tilt of 11°. This could generate PV electricity of about 6.67 TW h per annum. This paper is important as it could serve as first-hand scientific information for government agencies, NGOs, investors, researchers and policy makers at maximizing the potential of solar PV rooftop potentials in Nigeria.

1. Introduction

The increasing population growth as well as the quest for better quality of life are mainly responsible for growing demand for electricity generation. In the past, electricity generation from fossil fuels has been continuously exploited without any consequences [1], today climate change is a reality and had raised several questions about the exploration of fossil fuels for electricity generation [2]. Renewable Energy (RE) sources are now widely accepted as a substitute to fossil fuels in meeting the growing demand for electricity around the world owing to the quest to mitigate the health and the environmental challenges pose by electricity generation by fossil fuels. As the carbon-based fossils are becoming increasingly scarce, RE sources are becoming popular and are taken the centre-stage in policy decisions across the globe. Technology in the solar power industry is constantly evolving and improving especially with the advents of feed-in tariff (FIT) program and various government supports in many part of the world. These various incentives have led to the proliferation of solar PV panel technologies resulting into continuous crash in the cost of PV modules which has made solar PV more appealing and acceptable.

The global installed capacity for solar PV in 2016 stood at an equivalent hourly installation of 31,000 panels worldwide [3]. Solar PV is flexible, mature, possesses high learning rate and has one of its biggest advantages in its distribution of energy. It can be utilized in several diversities (kW, MW) and locations for both decentralised (industrial, residential and commercial) and utility scale (deserts, cities) grid-connected and off-grid systems [4,5]. Recent developments in the electric power industry have propelled the global acceptance of rooftop PV and building integrated photovoltaic (BIPV) systems in resolving the issues of land unavailability associated with cities [6,7]. Urbanisation and employment are two factors that have propelled rural-urban migration towards cities which in turn, have become the epicentre of economic development with its attendant challenges of greenhouse gas (GHG) emissions and energy unsustainability [8]. Globally, the

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Nomenclature and list of symbol

ρ_g	Ground reflectance (or albedo)
TC_η	Temperature coefficient of Power
T_C	PV cell temperature
G_{eff}	Incident solar radiation
ϕ	Latitude of the location
ω	Hour angle
δ	Angle of declination
α	Panel inclination angle
α_s	Solar altitude angle
θ_Z	Zenith angle
GHG	Greenhouse gas emissions
RE	Renewable energy
LiDAR	Light detection and ranging
GDP	Gross Domestic Product
PD	Population Density
GSU	Geographic sub-unit
γ	Azimuth correction angle
A_{PV}	Available roof area
$A_{roof/cap}$	Roof area per capita
PV_{suit}	Suitable photovoltaic area
PV	Photovoltaic
GWp	Peak gigawatts
PB	Payback period
LGA	Local Government Area
dc	Direct current
LCOE	Levelised Cost of Electricity
MW_p	Peak megawatts
GWh/year	Gigawatt-hour per year
GIS	Geographic Information System
DNI	Direct Normal Irradiance
FIT	Feed-in-Tariff
sq.m	Square metre
sq.km	Square kilometre

residential sector consumes about 40% of total energy consumption, and emits more than half the amount in GHG emissions [4]. With the rapid surge towards cities, electricity demand will increase uncontrollably causing inevitable energy shortage, unless efforts are made towards providing affordable and sustainable energy.

The estimation of rooftop PV potential on an urban landscape is rigorous as a result of shadow effects, varying building orientations and challenge of little or no data availability. It was generally believed that of all RE technologies, rooftop PV systems remain inconsistent as a result of changing urban architectural complexities [9]. While several work has been done around the world in estimating the potential of rooftop PV system, to the best knowledge of the authors non has been done in Nigeria. This can be a hindrance to the development of rooftop solar PV technology as no concrete scientific information is available for optimal investment in rooftop PV technology in the country. This paper therefore investigates the potential of rooftop solar PV system for electricity generation in Nigeria.

2. Review of literature

Over the years, many researchers have proposed different methods in identifying rooftop suitability and PV technical potential. They are broadly divided into three [10–12] which are: the constant value method, the manual selection technique and the GIS based methods. The constant value method involves the use of representative samples in providing a reliable estimate for available roof area which is then extrapolated to the total roof area. Most of the researchers have used rule-of-thumb assumptions in identifying rooftop suitability. The total roof area is determined by multiplying the roof area per capita to the total population of the area under study, and also involves correlation of population density and available roof area. It is mainly used in assessing large regional areas of rooftop PV potential. It is inexpensive and has its main setback in its difficulty in validation/accuracy owing to several nuances obstructions, shadings but they provide a precursor for rooftop PV generation and have been proven to show reliable estimate. Constant value method does not require strict rule as to how many representatives should be selected, however as a rule of thumb, it should not be less than 10% of the total GSUs. This method is not as accurate as the other two methods (i.e. the manual selection technique and the GIS based methods) but has the advantage that it can be used for a large area such as the one considered in this study. The method can serve as a first hand scientific information where no concrete scientific information exists. The Manual Selection methods offers a more accurate approach than the constant value method, albeit time-intensive. In this technique, each building is assessed in detail based on statistical data and individual building characteristics such as number of floors, ground floor areas, number of buildings, etc. It makes use of high-resolution satellite imagery or aerial photographs in evaluating rooftop PV potential. It is mainly utilized in smaller regions and needs a thorough understanding of the local assessment (study area). The third method is the GIS-based technique in which optimal values set for representative rooftop characteristics are input into a computer model and the GIS software thus identifying areas of high suitability. In ascertaining shadow effects or solar potential, 2D images or 3D models are employed; these 3D models are most times produced from photogrammetry, advanced cartographic dataset or LiDAR data and are integrated with slope, orientation and building topography. This method is quicker, easily automated and more accurate than the previous methods but is time and computer-resource intensive.

Many researchers have utilised either manual selection technique or GIS-based methods which employ the use of LiDAR data, statistical building datasets and high-resolution satellite imagery with a spatial resolution of 0.5 m to estimating the rooftop solar PV potential in many parts of the world. For example Hofierka and Kanuk [13] have presented a methodology for the assessment of PV potential using open-source solar radiation tools and a 3D model implemented in a GRASS-GIS environment. The PV potential was assessed using the PVGIS tool and the buildings categorization and topography was implemented using the GRASS-GIS software. Results showed that the high PV potential could satisfy two-third of the electricity consumption in eastern Slovakia. In another study, Ordonez et al. [14] proposed a methodology to determine the solar generation potential on rooftops in Andalusia, Spain. Representative sampled buildings were analysed using local building construction statistics according to their typologies: detached/semi-detached houses, townhouses/row houses and high-rise buildings. After roof reductions, 51–55% and 16–21% of flat and pitched roof surfaces respectively were available for PV installation and deployment. Results show that the annual generation potential derived was 9.73 GW h/year and corresponds to 78.89% of all local electric demands. Similarly, Bergamasco and Asinari [15] proposed a methodology for PV solar generation potential for the Piedmont region in Italy. This was accomplished using the hierarchical perspective through the evaluation of useful global solar radiation, available roof area and their performances. The GIS cartographic dataset was utilized in computing the total roof areas using GIS by means of the association of polygons, with a gross roof area of 24 sq km and 19 sq km for industrial and residential buildings respectively. Using different scenarios, an available energy of between 5.8 TW h/year and 6.9 TW h/year was determined, constituting about 28% of the local energy consumption. Brito et al. [16] proposed a procedure for estimating PV potential of an urban region, Lisbon from the LiDAR data using the Solar Analyst tool. The total PV potential obtained indicated an installed capacity of 7 MW for the sampled 538 buildings with a potential of about 11.5 GW h/year corresponding to about 48% of local electric demand. These results suggest a first-approximation estimation of the PV potential of an urban area without the need for a full 3D analysis of shading on a Digital Surface Model. Singh and Banarjee [17] proposed a method for estimating the macro-scale RFTPV potential for an area using the freely available Google Earth imagery with inputs from micro-level simulations. Micro-simulations for the rooftop system sizing and design were carried out using the *PVSyst simulation* software on different types of buildings. Results obtained gave generic factors which enable extrapolation using the GIS software, QGIS 1.8.0 to the entire study area, IIT Bombay campus with more prospects on expansion to macro-scale areas. Lukac et al. [18] proposed a novel approach in determining a rating list of roofs' surfaces within urban areas with respect to their solar potential and suitability for installing PV systems. This was carried out by merging extracted urban morphology from LiDAR data with the pyranometer measurements of diffused and global solar irradiances. A shadowing model was established indicting the shadow effects caused by vegetation and surrounding buildings and trees. Results showed an improvement in the accuracy of the proposed method, and a 97.4% correlation with on-site measurements measured from a grid-tie plant when compared. Yuan et al. [19] proposed a method to calculate the rooftop PV potential by estimating the available roof area and incident solar irradiation for the city of Osaka. They make use of an aerial photo data, manual selection of rooftops and using the pixel analysis with C++ program. The study area was divided in 24 1sq. km samples before been analysed based on individual building characteristics. Results show an available roof area of 42 ± 9sq.km and corresponds to 56% and 34% of the total electric demand for the commercial and commercial/industrial sectors respectively. Dehwah et al. [4] employs the use of RS and GIS techniques to investigate the solar potential on rooftops in the residential sector of the Kingdom of Saudi-Arabia (KSA). PV SOL software was used to determine the impact of shading and model the application of PV applications on rooftops. An available roof area of 18–25% was obtained, with an annual estimation potential of 588 GW h which can satisfy about 35% of local electric demand for residential buildings in KSA. Kumar et al. [20] have evaluated the performance of thin film BIPV as a double sloped pitched roof in buildings of Malaysia. In their work, the effect of varying roof pitch and orientation as well as energy generation and performance parameters were evaluated for pitched roof BIPV. The study revealed that as the roof pitch angle increases the performance of BIPV slightly reduced. The authors concluded that the performance of the BIPV inclined at 15° and East orientation was better among the other orientation and angles in Malaysia. Shukla et al. [21] have identified sustainable building concept in South Asian countries as well as the role of BIPV applications in sustainable building. The authors also highlighted the possible barriers and challenges of BIPV system as related to South Asia countries. It was concluded that shadow effect, ambient temperature, PV direction as well as the slope of the PV are the important factors to be considered to get higher power output and high efficiency in practical applications.

All the aforementioned literature utilised either manual selection technique or GIS-based methods which is believed to be inconsistent for the assessment of large areas [11,22]. They are also limited in terms of availability with high cost constraints. For this reason, many researchers have used constant value method which utilized the sample roof areas within irregular cities. For example, Lehmann and Peter [23] sought to find a mathematical description and interrelationship between roof (façade) area and population density in Europe, using statistical data obtained from residential and non-residential buildings in Northwine-Westfalia, Germany. They assume the data to be valid and extrapolate to the rest of Europe based on building and population densities. The statistical approach was to fit into curves between the population density (on the x-axis) and the roof (and façade) per capita. Results show a $4-9\ m^2$/cap available roof area and $3-5\ m^2$/cap for available façade area for non-residential buildings. However, a $4.5-7\ m^2$/cap and $5-6\ m^2$/cap area was obtained on residential buildings for available roof and facade areas respectively. Applying a factor of 0.9 for roof losses and 0.66 for reduction on façade areas due to shadowing, the result was extrapolated to the rest of Europe. Paidipati et al; Denholm and Margolis [24,25] estimate the available roof area using floor space data to extrapolate to the total roof area. Several assumptions were utilized on the mix of building types (i.e. flat or pitched roofs). Ladner-Garcia and Carrillo [26] applied a constant value to total building area to estimate the rooftop PV potential for Puerto-Rico. A typical square footage for all building typologies (residential, commercial, industrial) was calculated using census data from the US census database. Their result shows that about half of all roof areas are deemed suitable for PV deployment. They base their penetration limits on the energy requirements of each sector/typology; and claimed that 25% of residential rooftops will satisfy the energy requirements of the sector. Wiginton et al. [22] proposed a method to integrate the capabilities of GIS, census data and object-specific image recognition using an extension of ArcGIS, feature analyst to determine the available roof area for PV deployment. A five-step procedure involved geographical divisions of the region, sampling and then extrapolation to the total roof area using the population versus roof-area correlations as argued by [23]. A linear relationship was determined with a roof area per capita of $70\ m^2/cap \pm 6.2\%$. Results indicate an available roof area of $13.1\ m^2/cap \pm 6.2\%$, and a potential peak power output of 5.74 G W which corresponds to 5% of the local electric demands of Ontario. Vardimon [27] estimated the rooftop area in Israel using a GIS software and aerial orthophotos. The buildings were classified into residential, industrial and commercial categories and a constant value of 30% applied to the entire building within Israel. Reduction considerations were sought from literature and local solar installers using a worst-case scenario and a mean rooftop per capita of 10.2 sqm/cap. Two scenarios were created with utilization factors based on building typology. Results show that 32% of local electric demand was met using a total potential scenario while the economic scenario meets 7–10% of local demands using rooftops greater than 800sqm. Khan and Arsalan [28] proposed a six-step procedure, by integrating census data, GIS and ENVI EX software. The roof area-population relationships were explored and extrapolated to the total roof area with a linear relationship of 13sqm/cap for the DHA Phase 7 in Karachi. Reductions for shading, other uses and building orientations were used to calculate the available roof area. High-resolution satellite imagery was utilized for validation, with a peak power output of 12.24 MW satisfying about 12.24% of the local electric demands.

While the roof areas have been conducted in many areas of the world, there is paucity of literature on determination of effective roof area for PV application in Nigeria. Lack of technical scientific information could hinder development of rooftop PV system. Assessment of available roof area for rooftop PV application is site specific and should be conducted at every region of the world. In this paper, constant value method is adopted to determine the potential of rooftop solar PV in Ibadan, South Western Nigeria. Although the technique has its limitation, but is believed to provides a reliable estimate especially in large areas such as the city of Ibadan. Moreover, the information and data required by this method is available. The inter-row shading and panel inclination angle were taken into consideration in determining the available roof area for the city of Ibadan. The methodology was based on the available rooftop area, using Google Earth imagery and GIS tools. The methodology is step wisely reported to enhance its repeatability for deployment in other areas. The study could provide a baseline for investors, policy-makers and researchers in the area of energy in promoting effective implementation of achieving considerable amount of energy from the roof-top solar PV. The study is one of the first attempts at providing an assessment of the solar PV potential in Nigerian cities to the best of our knowledge with the aim of providing support for policy-makers and investors in realizing optimal investment in roof top PV in Nigeria.

3. Overview of the study area

Nigeria is Africa's most significant country in terms of population with an estimated 180 million people covering a total surface area of 923,768 km^2 as of 2017. It lies between the latitudes of $4°18'54''N$ and $13°43'45''N$, then longitudes of $2°43'60''E$ and $14^{\circ}13'2''E$, thus described as a ''low-latitude'' region. As one of the fastest growing economies of the world, its energy consumption per capita is about 144 W in 2014, compared to Angola's 312 W with a population that is 82% lesser than Nigeria. The country can scarcely meet the demand of her citizenry with an available capacity of about 5000 MW especially in rural areas where about 51% of her population reside [29]. A cosmopolitan city of Ibadan was chosen for this study as a result of its high insolation, suitability for PV modules installation, and ever-increasing energy demand. In view of its latitudinal location, it has high solar irradiation and sunlight hours with an estimated average global solar radiation of 4.616 kW h/m^2/day [30]. It lies on the geographical co-ordinates of longitude $3.54°E$ of the Greenwich Meridian and latitude $7.38°N$ of the equator. Ibadan is the capital city of Oyo state and occupies a land area of about 3213.30 sq.km [31]. The city of Ibadan is the third most populated city in Nigeria and is the largest city by landmass in West Africa. The city comprises of eleven (11) local government areas (LGAs) i.e. municipalities as depicted in Figure 1 which are primarily residential, small-scale commercial buildings with a few industrial buildings; it has an overall population density of 985 persons per sq. km, majority of whom are traders and are involved in micro-businesses [32].

4. Methodology

The availability of data and provision of high-resolution satellite imagery is herculean and inaccessible in Nigeria. This has prompted the use of constant-value methods and Google Earth satellite imagery in this study which is freely accessible. The study was conducted in two phases: In the first phase, GIS was employed to identify and digitize rooftops within the Geographical Sub-Units (GSUs), and in the second phase, the annual generation potential was determined. A hierarchical procedure as itemised in Ref. [33] was adopted and implemented in the estimation of rooftop PV potential of the study area. The overall workflow and techniques associated with the determination of the available rooftop area is illustrated in Figure 2. Firstly, the optimal tilt angle and solar irradiation on tilted surfaces were determined using mathematical models of the horizontal solar irradiance data. Secondly, the Google Earth satellite imagery of the study area was segmented into smaller sub-units using the ward shape-file. This is done to ensure that land area and population density data can be easily extracted from the Geographical sub-units (GSUs) that make up the area under study.

The information extracted from the GSUs were utilized in such a way that the roof area for 24 of the 123 GSUs was extracted by manual digitization and geo-referencing in the ArcGIS environment. Thereafter, the randomly sampled roof areas were then extrapolated first to the Local government areas (municipalities) and then to the entire targeted area, in order to ascertain the total roof area, and thus, the annual energy output.

4.1. Determination of optimal tilt angle

The estimation of solar irradiation on tilted surfaces is important for PV deployment and increased energy output. Optimum tilt angle is site-specific since it depends on the latitude of the location, climatic variables and position of the sun [34]. Solar PV modules therefore, should be installed where there is maximum insolation, good orientation and at optimum tilt devoid of shading for maximum extractable solar power. While global solar irradiations on horizontal plane are readily available by measurement, solar irradiation on tilted surface are not available in Nigeria, however, it could be determined from the measured values of global solar irradiation on horizontal surfaces. Therefore, the daily optimal solar irradiance on tilted surfaces in this paper were determined using the available daily average horizontal global solar radiation data obtained from the International Institute of Tropical Agriculture, Ibadan over the periods of 2008–2018.

FIGURE 1
Map describing the study area.

The average daily total solar radiation incident on a tilted surface, H_T (Wm^{-2}) can be estimated as follows [35]:

$$H_T = H_B + H_D + H_R \tag{1}$$

where, H_B is the daily beam solar radiation incidence on a tilted surface, Wm^{-2}; H_D is the daily diffuse solar radiation incidence on a tilted surface, Wm^{-2}; and H_R is the daily reflected solar radiation on a tilted surface, Wm^{-2}. Their values can be calculated using the following relations [35]:

$$H_B = H_b \times R_b \tag{2}$$

$$H_D = H_d \times R_d \tag{3}$$

$$H_R = \left(\frac{1-\cos(\alpha)}{2}\right) \times \rho_g H \tag{4}$$

Where ρ_g is the ground reflectance or albedo (=0.2); α is the panel optimum tilt angle, H is the horizontal global irradiation obtained from the meteorological station, H_b and H_d are the beam and diffuse radiation components irradiating on a horizontal surface, and they can be evaluated using the Page model as follows [36]:

$$H_b = H - H_d \tag{5}$$

$$H_d = (1 - 1.13 \times K_T) \times H \tag{6}$$

K_T is the clearness index and is represented by;

$$K_T = \frac{H}{H_o} \tag{7}$$

where, H_o ($kJm^{-2}day^{-1}$) is the average daily extra-terrestrial radiation falling on the horizontal surface, and can be determined using [37] :

$$H_0 = \left(\frac{24}{\pi} \times 3.6 \times I_{SC}\right)\left[1 + 0.033\cos\left(\frac{360 \times m}{365}\right)\right] \times \left(\cos\phi\cos\delta\sin w_s + \left(\frac{2\pi}{360} \times w_s \times \sin\delta\sin\phi\right)\right) \tag{8}$$

where, I_{SC} is the solar constant (1367 Wm^{-2}), ϕ is the latitude of the location and m is the Julian day of the year; w_s is the sunrise hour angle and δ is the angle of declination.

The δ and w_s can be determined as follows:

$$\delta = 23.45 \times \sin\left[360 \times \left(\frac{m+284}{365}\right)\right] \tag{9}$$

$$\cos w_s = -\tan\delta \times \tan\phi \tag{10}$$

R_b and R_d are the transmittance-absorptances of the beam and diffuse components, respectively. They are dependent on the local climate, cloudiness and concentration in the atmosphere. The transmittance-absorptance of the beam component is determined

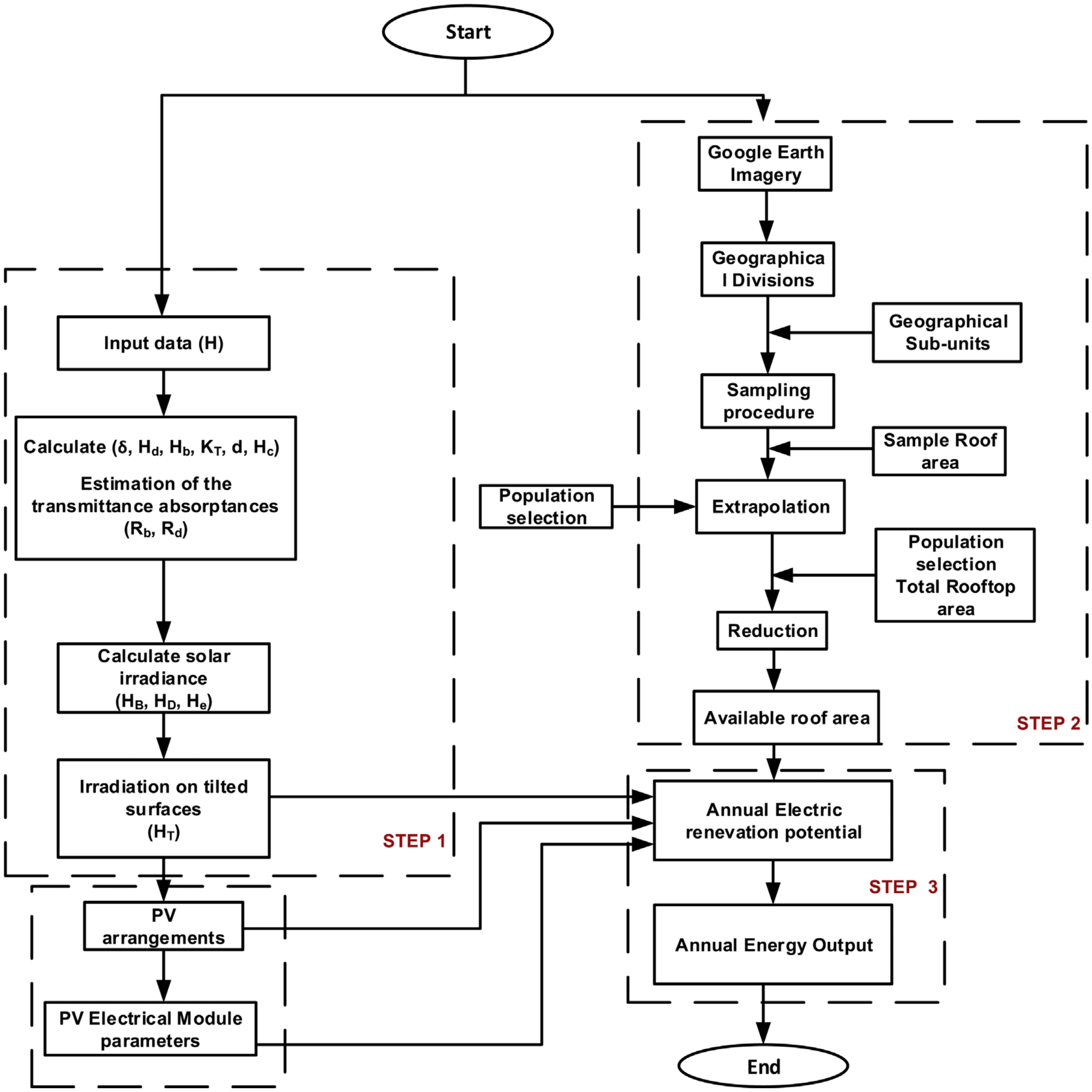

FIGURE 2

Methodology framework of the study.

using Liu and Jordan model [38] as follows:

$$R_b = \frac{\cos(\phi-\alpha)\cos(\delta)\sin(w) + w\left(\frac{\pi}{180}\right)\sin(\phi-\alpha)\sin(\delta)}{\cos(\phi)\cos(\delta)\sin(w_s) + \left(\frac{2\pi w_s}{360}\right)\sin(\phi)\sin(\delta)} \tag{11}$$

α represents the optimal tilt angle, and w is the sunset hour angle and estimated by:

$$w = \min\{w_s, \cos^{-1}[-\tan(\phi-\alpha)\tan\delta]\} \tag{12}$$

While the transmittance-absorptance of the diffused component can be calculated using:

$$R_d = \left((1-A_i)\left(\frac{1+\cos\alpha}{2}\right)\left[1+f\sin^3\left(\frac{\alpha}{2}\right)\right] + A_i R_b\right) \tag{13}$$

where, f is the horizontal brightening component and A_i is the Anisotropic sky. Both

f and A_i can be calculated using the following relations:

given as:

$$f = \sqrt{\frac{H_b}{H}} \tag{14}$$

$$A_i = \frac{H_b}{H_0} \tag{15}$$

4.2. Determination of available roof area

This section presents the methodology used to determine the available roof area with the aim of evaluating the potential of extractible solar PV energy for the city of Ibadan.

4.2.1. Divisions into geographical sub-units

To carry out the analysis of rooftop area, it is important to partition the study area into smaller units referred to as the Geographical

Sub-units (GSUs). These sub-units constitute to serve as the smallest administrative boundary, thus representing varying amounts of rooftops across several localities. In this study, Wards (i.e. a smaller unit of municipality within the study area) were used as GSUs so as to achieve better accuracy. This technique was employed in Refs. [22,28] where Region of Interests (ROIs) were divided into smaller administrative census sub-divisions in Karachi and Ontario, respectively. Based on the non-availability of data, ward shape-files created for the study region were fully hand-digitized into the ArcGIS environment in order to obtain the administrative units. In all, there are 123 non-overlapping GSUs within the city of Ibadan which characterize the region under study.

4.2.2. Sample selection

The population size of buildings in the 123 GSUs were too many for practical determination of rooftop area, therefore, 24 of the 123 wards which serve as GSUs were randomly sampled. The area per hectare, the population of each of the randomly selected wards and ward classification data were obtained from the National Population Commission (NPC) and independent National Electoral Commission (INEC), respectively. This was to ensure the determination of the population density (PD) for each of the wards in order to obtain the population-roof area relationship. The PD is the ratio of the number of persons to the urban surface area, expressed in persons/hectare. Many studies [22,23,28,39–42] have identified a strong relationship between roof area and the population in their respective regions of interests, thus the 24 selected GSUs were classified into three different categories of low population density (0–100 persons/hectare), medium (101–500 persons/hectare) and high population density (>500persons/hectare). Of these selected GSUs, 6 were of low population density, 11 were identified as medium population dense areas while 7 were found to be of high population density. The population data were that of 1991 census when the ward census was last conducted in the country. Hence, the population was projected to the year under study using:

$$p_{(i)} = P(1+\frac{r}{100})^n \quad (16)$$

where p is the population been projected, P is the population as at 1991 (i.e. the base population), r is the growth rate which is 2.3% for Nigeria, i is the index for identifying the ward the population is being projected and n is the no of years.

The 24 selected GSUs, their population density, and type of settlement are furnished in Table 1.

4.2.3. Roof print area

Rooftops within the 24 selected GSUs were manually-digitized in the ArcGIS environment using Google Earth satellite imagery in order to obtain roof print area for each of the 24 wards. ArcGIS 10.4 was used in geo-referencing the image, query and run survey report in order to obtain the roof print area as utilized by Ref. [44] in Lahore. Figure 3 depict a sample representative of a digitized roof area for Mokola ward (ward 10). This was done for each of the wards to obtain roof print area.

4.2.4. Determination of roof area per capita

The digitization and extraction of roof area has been performed for each of the 24 selected wards using ArcGIS. This is very important so as to obtain the roof area per capita for each of the 24 selected

TABLE 1

24 selected GSUs, projected population densities and settlement classification [43].

Index (i)	Ward	Area(hectares)	Population	PD(cap/hect)	PD classification	Sectorial classification
1	Iwo Road	340.20	85,028	249.93	Medium	R, C
2	Eleyele	492.67	38,725	78.60	Low	R, C
3	Ekotedo	21.150	22,016	1040.95	High	R, C
4	Asukuna	10.27	23,565	2293.61	High	RR, C, A
5	Alekuso	10.66	2109	197.82	Medium	R, C
6	Ikolaba/Ashi	1036.41	34,011	32.82	Low	R, C
7	Igosun/Itutaba	50.03	30,437	608.33	High	RR, C, A
8	Yemetu/Adeoyo	199.90	78,962	395.003	Medium	R, C
9	Ode-olo/Oniyanrin	110.40	48,476	439.08	Medium	RR, C
10	Mokola	87.15	32,813	376.49	Medium	R, C, IN
11	Univ of Ibadan/Poly	1067.52	19,463	18.23	Low	R, C, I
12	Oke-Itunnu	189.89	40,981	215.82	Medium	R, C
13	Sabo	36.85	14,432	391.65	Medium	R, C
14	Ring Road	1766.895	145,655	82.44	Low	R, C
15	Molete	244.33	10,208	41.78	Low	R, C
16	Isale-Osi/Gege	40.16	44,040	1096.58	High	R, C, A
17	Ogunpa	77.96	10,732	137.67	Medium	RR, C
18	Popo Yemoja	66.91	41,792	624.60	High	RR, C
19	Iyaganku	332.03	27,652	83.28	Low	R, C
20	Oke-Ado	182.77	123,716	676.91	High	R, C
21	Isale-Ijebu	30.99	6719	216.793	Medium	
22	Foko	38.26	60,528	1582.23	High	RR, C, A
23	Aperin/Gbelekale	132.49	40,934	308.97	Medium	R, C
24	Agugu/Koloko	195.43	90,275	461.92	Medium	RR, C

[RR]Rural Residential [C] Commercial [A] Agricultural[R] Residential [I] Institutional [F] Industrial.

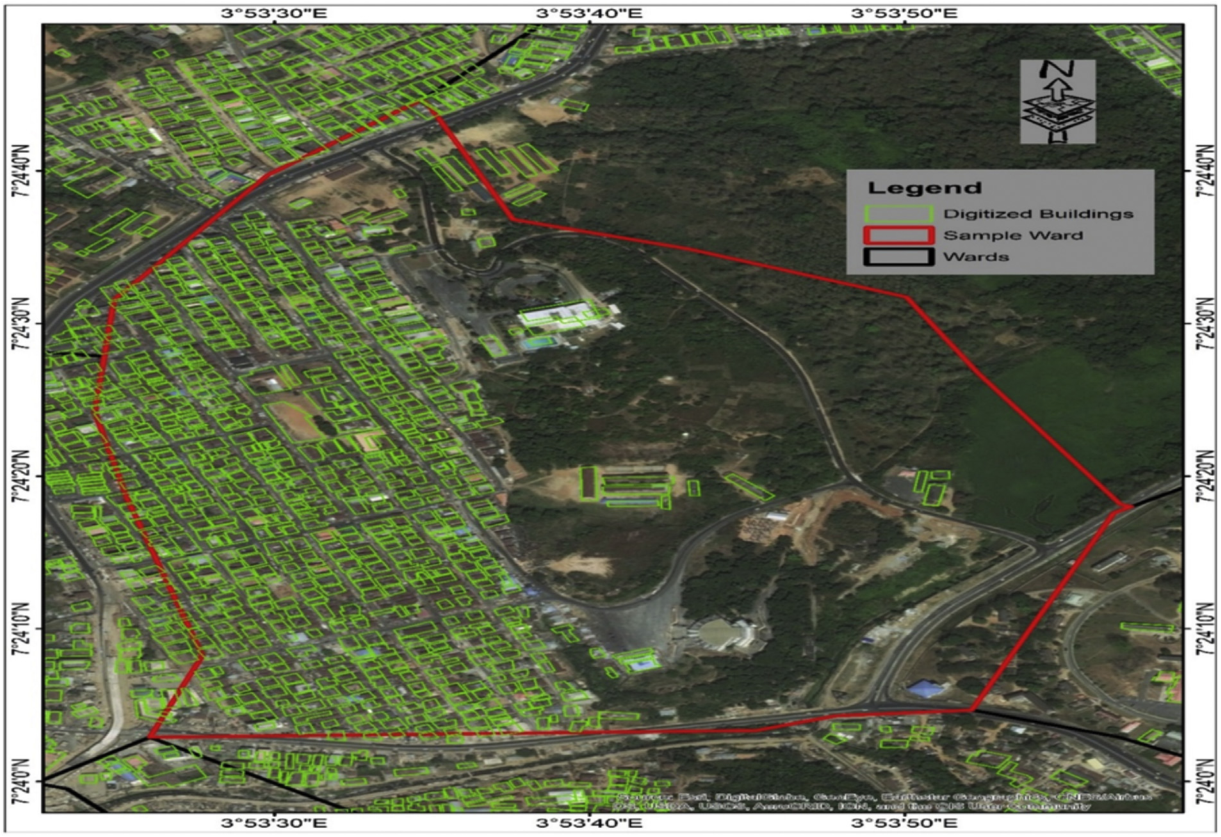

FIGURE 3
Representative sample of digitized building rooftops (in ArcGIS) (Mokola ward).

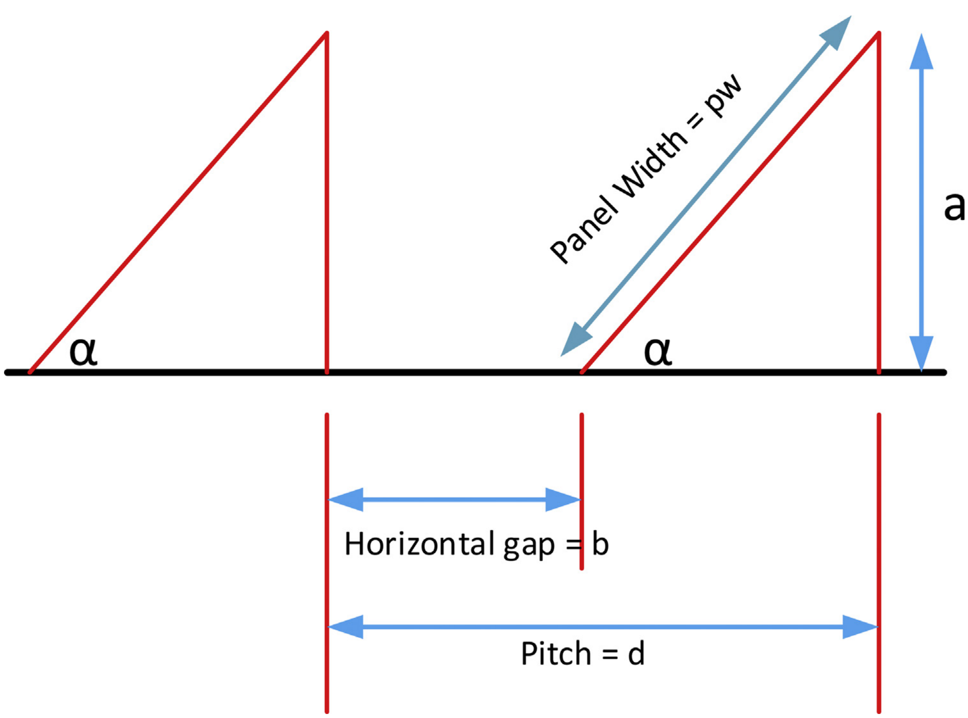

FIGURE 4
Sketch showing the relationship between the module array and the reserved distance to prevent panel-to-panel shading.

representative wards. The roof area per capita for each ward (GSU) was determined as follows:

$$A_{roof/cap}(i) = \frac{A_{(i)}}{p_{(i)}} \quad (17)$$

where $A_{(i)}$ is the roof print area value for the selected ward i obtained from ArcGIS as obtained in Section 4.2.3, $p_{(i)}$ is the present population of the ward i as obtained from NPC.

4.2.5. Extrapolation to study area

It is impractical to manually-digitized all the GSUs in the ArcGIS, hence the need for the representative. Having determined the roof area per capita for each of the representative $A_{roof/cap}(i)$, the average roof area per capita for the entire study area was determined using the mean of the representative of the roof area per capita as follows:

$$AVE_{roof/cap} = \frac{\int_{i=1}^{24} A_{roof/cap}(i)}{N = 24} \quad (18)$$

It should be noted that once the average roof area per capita is determined, it can be used to determine the total roof area of any given location within the study area once the population is known. Hence, the important need to determine the average roof area per capita.

Therefore, the total roof area of any location within the study area can be determined using:

$$A_{roof} = AVE_{roof/cap} \times p \quad (19)$$

where $AVE_{roof/cap}$ is the average roof area per capita as determined using (18) and p is the population of any location within the study area

4.3. Determination of available roof area for solar PV deployment

In order to obtain available roof area that is suitable for determining the potential extractable energy from solar PV, there is need to reduce the total roof area based on the factors that influence fractional availability of roof areas. Therefore, the total roof area determined in (19) is further reduced due to (i) shadowing effects from neighbouring trees, buildings, and other elements on the rooftop (ii) architectural elements on the roof such as water storage tank, HVACs, pipeline vents, etc., (iii) shadings obtained from the installation and tilt of the PV panel itself (iv) inclination of the surface horizontal where PV is mounted and (v) orientation of the rooftop.

4.3.1. Reduction by building orientation, construction restrictions

Building orientation is important in the deployment of PV applications. Many studies have used general assumptions by employing a constant value across all building stocks [26] or based on the building type [24]. Field Surveys indicate the study area to be a thorough residential neighbourhood with predominant usage of traditional buildings and peaked roofs. Most buildings in Ibadan (the study area) been peaked rooftops are used for mixed purposes, both residential and commercial activities constituting about 55% of the buildings, hence $\mathbf{W}peaked$ = 0.55; and 16% for flat rooftops, $\mathbf{W}flat$ = 0.16 [45]. Buildings with peaked rooftops tend to be lower in elevation, slant and are less exposed to shading. They are affected by orientation with 50% south-facing area ($\mathbf{T}peaked$ = 0.50). Furthermore, for services and industrial buildings which are most times high-rise with flat (or monitor) rooftops; they are unaffected by building orientations, $\mathbf{T}flat$ = 1.

Thus, considering the approach of [28], the fraction of properly oriented rooftop (ROF) becomes:

$$ROF = (T_{peaked} \times W_{peaked}) + (T_{flat} \times W_{flat}) \quad (20)$$

4.3.2. Reduction by shading by other buildings

The factors of building orientation and unshaded areas yield the utilizable fraction of the available roof area necessary for PV deployment. This correlates the effect of shadows, other uses on the rooftop, PV installation, servicing and maintenance. Assessing the values of shadowing effects depends on the building type and country, using panchromatic or hi-spec data imagery involving 3D models. Several researchers have used different methods in estimating the fraction of available roof area due to shading and other roof uses ranging between 0.30–0.35 [22,28,40,46]. The best-case scenario of roof reductions, affected by shading, construction restrictions and other roof uses ($H_{shading}$) has been used in this research and is taken as [22,28]:

$$H_{shading} = 0.35 \quad (21)$$

Thus, the available roof area (A_{pv}) for PV deployment which is the total roof area (A_{roof}) reduced by building orientation, construction restrictions and shading factors can be determined as follows:

$$A_{pv} = A_{roof} \times H_{shading} \times ROF \quad (22)$$

4.4. Inter-row shading of rooftop photovoltaic systems

This is caused by one row of tilted PV modules casting a shadow on another, thereby reducing PV performance and efficiency. Solar irradiation placed horizontally possess the capability of an estimated 8% loss in efficiency [47]. The shading of a row of PV modules on another reduces system performance and energy output and should be considered. This is more predominant in commercial and services buildings were roof space is limited, but must be accommodated for maximum ground ratio. Inter-panel shading constitute one of the biggest setbacks in the estimation of rooftop PV. One way of correcting this could be by placing panels horizontally. This cannot be upheld as a result of its lower energy output and dirt effects. Toikka [48] also describes a remarkable reduction of 70% in module efficiency when shaded by 2%, hence enforcing the need for preventing inter row shading. Inter row spacing is dependent on the panel orientation, latitude and PV module type/dimensions. Byrne et al. [12] estimated a 48% available roof area for the City of Seoul, with a module array tilt of 22°.

Thus, to determine the shading distance between the front and back rows, a reference PV module (SW280 mono black c-Si PV module) whose parameters are listed in Table 2 was utilised for this study.

Figure 4 describes the relationship between the module array and the reserved distance to prevent panel-to-panel shading. The

TABLE 2

Electrical Parameters for SW280 mono black c-Si PV module [49].

Parameters	Units	Rated values
Maximum Power, $P_{\max}$	W_p	280
Maximum Power Current, I_{mp}	A	9.07
Maximum Power Voltage, V_{mp}	V	31.2
Open circuit Voltage, V_{OC}	V	39.5
Short Circuit Current, I_{SC}	A	9.71
Number of cells		(6 × 10)
Module Efficiency, η_{mod}	%	16.70
Dimensions (t×1×w)	mm	33×1675×1001
NOCT	°C	47.5
Temperature Coefficient of Power, TC_η	%/°C	−0.43

At STC, Cell temp = 25 °C, Irradiance = 1000W/m^2, AM = 1.5G spectrum.

distance between the front and back rows of the PV modules are also illustrated in the figure.

where α is the optimal tilted angle as determined in Section 4.1, b is the horizontal gap between rows, a is the height of PV panel, pw is the PV module width and d is the pitch.

In calculating the inter-row panel shading, it can be seen that the PV module is installed at a fixed optimum tilt and a panel width (pw). The height difference is determined by calculating the distance from the slanted top of the PV module to its horizontal, using the Duffie and Beckman [50] approach. Therefore, the height of PV panel, a, as well as the horizontal gap, b, between the PV module when optimally tilted at a fixed point, α, between rows can be determined as follows:

$$a = \sin(\alpha) \times pw \tag{23}$$

$$b = N \times \cos\gamma \tag{24}$$

where, N is the solar elevation angle, γ is the azimuth correction angle, ϕ is the latitude of the location, δ is the solar declination, ω is the winter solstice hour angle (i.e. between 9:00am to 12:00pm (or 12:00−3:00pm), [50] at 15 every hour on December 21 st).

The solar elevation angle can be determined using:

$$N = \left(\frac{\alpha}{\tan\alpha_s}\right) \tag{25}$$

α_s is the solar altitude angle and is a function of the time of day and the declination angle and can be calculated using:

$$\sin\alpha_s = \cos\phi\cos\delta\cos\omega + \sin\delta\sin\phi \tag{26}$$

Similarly, the azimuth correction angle, γ can be determined using the following relations:

$$\gamma = sign(\omega)|\left(\cos^{-1}\frac{(\cos\theta_z \times \sin\phi) - \sin\delta}{\sin\theta_z \times \cos\phi}\right)| \tag{27}$$

$$\cos\theta_z = (\cos\phi \times \cos\omega \times \cos\delta) + (\sin\phi \times \sin\delta) \tag{28}$$

$$d = (\cos\alpha * p) + b \tag{29}$$

where θ_z is the zenith angle.

Note; $\cos\theta_z = \sin\alpha_s$.

4.5. Maximum installable capacity and annual PV energy production

The maximum installable PV capacity can be determined using the following relation

$$MW_p = \eta_{mod} * 1kW/m^2 * A_{PV} \tag{30}$$

While the total annual energy output E_{PV} in $GWh/year$ over a large scale rooftop PV deployment generated on tilted surfaces can be estimated using [26,51]:

$$E_{PV} = \frac{H_{T,mean} \times A_{PV} \times \eta_{inv} \times \eta_{PV} \times \eta_{mismatch} \times \eta_{dirt} \times 365}{10^6} \tag{31}$$

where, E_{PV} is the annual energy output of the solar PV ($GWh/year$), $H_{T,mean}$ is the daily mean average solar radiation on tilted surfaces ($kWh.m^2_{-}.day^{-1}$) as determined using (1)–(15) in Section 4.1,A_{PV} is the Available rooftop area (m^2) as determined using (22) in Section 4.3.2,A_{PV} is the efficiency of the PV module to be used, η_{inv} is the inverter efficiency, η_{dirt} is the losses due to dirt and $\eta_{mismatch}$ is the loss due to mismatch efficiencies. In determining the annual energy output in this paper, typical panel characteristics was employed (i.e. SW280 mono black c-Si PV module) with a given values of mismatch, dirt and DC to AC power inverter efficiencies and temperature variations under higher temperatures. Singh and Banerjee [17] have estimated the typical PV module to have η_{inv}, η_{dirt} and $\eta_{mismatch}$ efficiencies to be 0.90, 0.96 and 0.98, respectively and these values are adopted in this study.

The efficiency of a solar panel (η_{PV}) depends on the amount of solar irradiation it can amass (η_{mod}) as well as the impact of ambient temperature of its surroundings (M_{temp}). Therefore η_{PV} can be determined using:

$$\eta_{PV} = \eta_{mod} \times M_{temp} \tag{32}$$

Singh and Banerjee [51] indicated that the temperature can be quantified using the percentage change in the output of the panel when it deviates beyond the standard temperature of 25 °C. Therefore, M_{temp} can be evaluated using the following relations:

$$M_{temp} = 1 + 0.01TC_\eta(T_c - T_j) \tag{33}$$

$$(T_C - T_a)^O C = \left(\frac{(NOCT - 20)^o C}{800}\right) G_{eff} \tag{34}$$

where η_{mod} is the rated efficiency of the PV module (%), TC_η is the Temperature coefficient of power (%/^{O}C), T_j is the PV module temperature at STC (25 °C), T_C is the PV cell temperature (°C), T_a is the average value of ambient temperature (°C) and is taking as 26.8 °C for the city of Ibadan, $NOCT$ is the Nominal Operating Cell Temperature, G_{eff} is the Incident solar irradiation (Wm^{-2}) which ranges between 800 – 1000Wm^{-2}.

5. Results and discussion

This section presents the results in accordance to the methodology framework presented in Figure 2. The optimum tilt angle, the average roof area per capita, total roof area as well as the available roof area for rooftop PV deployment for the city Ibadan are presented in the preceding subsections. Also, the inter-row shading of rooftop photovoltaic systems, the maximum installable rooftop PV capacity as well as the probable energy available are also presented in this section.

TABLE 3

Monthly average solar radiation on tilted surfaces ($H_{T,mean}$) in Ibadan (MJm^{-2}day^{-1}).

Tilt Angle								
Month	**Horizontal surface (0°)**	**5.45..^{0}**	**11.00..°**	**16.36..^{0}**	**21.82..^{0}**	**27.27..^{0}**	**32.72..^{0}**	**38.18..^{0}**
Jan	12.73	13.90	14.33	14.66	14.91	15.07	15.14	15.13
Feb	14.87	15.50	15.77	15.96	16.05	16.05	15.96	15.78
Mar	14.97	16.94	16.98	16.91	16.74	16.47	16.11	15.66
Apr	14.5	16.52	16.24	15.87	15.41	14.87	14.25	13.57
May	14.04	15.50	15.02	14.46	13.83	13.14	12.40	11.61
Jun	12.60	14.16	13.65	13.08	12.45	11.78	11.07	10.33
Jul	9.78	11.64	11.32	10.95	10.54	10.09	9.602	9.092
Aug	9.03	10.58	10.40	10.18	9.912	9.604	9.261	8.886
Sep	11.17	12.80	12.73	12.60	12.40	12.14	11.82	11.45
Oct	13.18	14.46	14.63	14.71	14.71	14.63	14.47	14.23
Nov	14.54	15.35	15.81	16.17	16.42	16.58	16.64	16.59
Dec	13.15	14.30	14.81	15.23	15.56	15.80	15.94	15.98
Yearly Average	12.88	14.30	14.31	14.23	14.08	13.85	13.56	13.19

5.1. Optimum tilt angle for the city of Ibadan

The annual solar irradiation incidence on tilted surfaces within the study area was investigated by comparing the solar radiation on horizontal surfaces to that placed on surfaces at a fixed optimal tilt angle for the city of Ibadan. To achieve this, a MATLAB program was developed based on the mathematical models in Eqs. (1–15) and the results showing average solar irradiation on PV modules on different tilt angles throughout the year are presented in Table 3. The table reveals that the annual solar radiation on horizontal surface was determined to be 12.88 MJ m^{-2}day^{-1}.

A glance at the table to compare the solar radiation on horizontal surface to that of inclined surface reveals that there is a slight increase of solar irradiation on tilted surfaces compared to the horizontal surfaces (**0**°), hence more electrical power could be obtained from the solar panels if tilted round the year. The optimum tilt angle for the study area can be obtained when the panel is positioned between 6–11 degrees corresponding to an annual average solar radiation of 14.30–14.31 MJ m^{-2}day^{-1}. This will involve the use of trackers mounted side-by-side against the PV panel. Solar trackers have a peculiar drawback in its maintenance, mechanical capability and capital costs. Nonetheless, most installers sometimes use an ‘adjustable’ which has its own peculiar issues. Hence, it is of utmost importance that the PV modules be installed at a fixed optimal tilt angle and in this case at 11 degrees. Table 3 also revealed that solar radiation varies from month to month with the month of March being the month that exhibits highest solar irradiance. This is because the month of March is the peak of dry season in the study area. Similarly, the month of August exhibits the lowest solar irradiation, because it is the peak of raining season with extended cloud cover.

5.2. Determination of average roof area per capita for the city of Ibadan

In order to determine the total roof area of a given location for solar PV deployment, there is need to determine its average roof area per capita as this is what would be used to extrapolate the total roof area of any given location within the study area. To achieve this, hand digitization and extraction of roof print area was performed for each of the 24 selected wards using ArcGIS as explained in Sections 4.2.3 and 4.2.4 while the roof areas per capita for each of the selected ward ware determined using (17). Thereafter, the average roof area per capita for the study area was determined using (18). The results of the projected population as determined using (16), the roof print area and the roof area per capita for each of the selected 24 GSUs are presented in Table 4.

The table reveals that the low and medium-dense areas range between 12–74 m^2/cap and 10–19 m^2/cap, respectively. The highest roof area per capita was found to be 73.97 m^2/cap (Ikolaba/Ashi ward) while the lowest was 0.9597 m^2/cap (Isale-Osi/Gege ward). The average roof area per capita for the entire study area was determined to be 14.35 m^2/cap using (18). An insight into previous

TABLE 4

projected population, roof print area and roof area per capita of the 24 selected GSUs.

	Ward	Projected population	Roof print area (sq m)	Roof area/cap (sq m/cap)
1	Iwo Road	85,028	909907.1718	10.70126513
2	Eleyele	38,725	568262.297	14.67430076
3	Ekotedo	22,016	46247.09054	2.100612761
4	Asukuna	23,565	33447.88426	1.419388256
5	Alekuso	2109	38498.45009	18.2543623
6	Ikolaba/Ashi	34,011	2,515,644	73.96559936
7	Igosun/Itutaba	30,437	111577.1132	3.665838066
8	Yemetu/Adeoyo	78,962	467637.7519	5.922313921
9	Ode-olo/Oniyanrin	48,476	279183.2339	5.759205254
10	Mokola	32,813	160442.9432	4.889615188
11	Univ of Ibadan/Poly	19,463	706456.2643	36.29739836
12	Oke-Itunnu	40,981	577567.5285	14.09354404
13	Sabo	14,432	85336.40133	5.912998983
14	Ring Road	145,655	1681643.472	11.54538788
15	Molete	10,208	672254.616	65.85566379
16	Isale-Osi/Gege	44,040	41596.0662	0.944506499
17	Ogunpa	10,732	164197.5196	15.29980615
18	Popo Yemoja	41,792	260250.8253	6.227288124
19	Iyaganku	27,652	451135.4712	16.31475015
20	Oke-Ado	123,716	399212.583	3.226846835
21	Isale-Ijebu	6719	63663.46719	9.475140227
22	Foko	60,528	119228.734	1.969811228
23	Aperin/Gbelekale	40,934	423324.8414	10.34164365
24	Agugu/Koloko	90,275	501097.3244	5.55078731

TABLE 5

The available roof area through the reduction of the total rooftop area.

LGA making up the city of Ibadan	Base population (1991 census)	Projected population to 2020	Total roof area (A_{roof}) $A_{roof} = AVE_{roof/cap} \times p$		Available roof area (A_{pv}) $A_{pv} = A_{roof} \times H_{shading} \times ROF$	
			(m^2)	~km^2	(m^2)	~km^2
Akinyele	140,118	271,060	3,888,100	3.888	592,000	0.592
Egbeda	129,461	250,340	3,592,400	3.592	546,900	0.547
Ibadan North (ibN)	302,271	584,510	8,387,700	8.388	1,277,000	1.277
Ibadan North East (ibNE)	275,627	532,990	7,648,300	7.648	1,164,500	1.164
Ibadan North West (ibNW)	147,918	286,030	4,104,600	4.105	624,900	0.625
Ibadan South East (ibSE)	225,800	436,630	6,265,700	6.266	954,000	0.954
Ibadan South West (ibSW)	227,047	439,050	6,300,300	6.300	959,200	0.959
Ido	53,582	103,610	1,486,800	1.487	226,400	0.226
Lagelu	68,901	132,240	1,911,900	1.912	291,100	0.291
Oluyole	91,527	176,990	2,539,800	2.540	386,700	0.387
Ona-Ara	123,048	237,940	3,414,400	3.414	519,800	0.520
Total (Ibadan)	1,785,300	3,452,300	49,540,000	49.540	7,542,500	7.543

research initiatives shows the average roof area/capita of 14.35 m^2/cap is a reasonable estimate for the city of Ibadan. Izquierdo et al. [40] reported a total roof area/cap of 41.7 m^2/cap for the whole of Spain, Wiginton et al. [22] revealed roof area/cap of 70 m^2/cap for Ontario; Khan and Arsalan [28] indicated 13 m^2/cap for Karachi; Ordonez et al. [14] reported 32.3 m^2/cap for Andalusia; Pratt [42] indicated that the roof area/cap for United Kingdomm is 10.6–30.7 m^2/cap; 74.4 m^2/cap for Sweden [52] and 17.6–21.2 m^2/cap for roof rain-water harvesting in Brazil [53].

5.3. Available roof area of the study area

The available roof area of the city of Ibadan was clearly determined by applying the varying reduction constraints enumerated in Section 4.3 to the total roof area and the results are presented in Table 5. The table indicates that the urban centres of ibN, ibNE, ibSW, ibSE and ibNW have the highest available roof area for roof top-PV deployment with the values of 1.3, 1.2, 0.96, 0.95 and 0.63 km^2, respectively despite the low land mass area compared to peri-urban centers. This is due to the huge population density leading to high number of buildings available for PV deployments. Ido, Lagelu and Oluyole municipalities have the least available areas for roof top PV deployment with the value of 0.226, 0.291 and 0.387 km^2, respectively despite their large land mass. The total roof area for the city of Ibadan is estimated as 49.54 km^2 while the available roof area for PV deployment stands at 7.54 km^2. This indicates that the available PV roof areas represents about 15.23% of the total roof area denoting a utilisation factor of 0.15.

5.4. Inter-row shading of rooftop photovoltaic systems for the city of Ibadan

In determining the horizontal gap between PV modules for the city of Ibadan to prevent panel-to-panel shading, using the reference panel characteristics depicted in Table 2 (i.e. SW280 mono black c-Si PV module) and Eqs. (23)– (29), the results are presented in Table 6. The table reveals the estimated values required to prevent inter-panel shading for roof top PV application in the city of Ibadan. The solar panel width (*pw*) was estimated to be 1001 mm while the panel height (*a*) was calculated to be 191 mm. The inter-panel horizontal gap (*b*) that would prevent panel-to-panel shading was estimated to be 154 mm with the pitch (*d*) value of 1137 mm.

TABLE 6

Results of inter-panel shading to prevent panel-to-panel shading.

Parameters	Values
Optimum tilted angle, α	11°
Panel Width, *pw*	1001 mm
Height difference, *a*	191 mm
Solar declination, δ	−23.45°
Hour angle, ω	45°
Latitude, ϕ	7.38°
Zenith angle, θ_z	53.69°
Solar altitude angle, α_s	36.31°
Solar elevation angle, N	260mm
Azimuth correction angle, γ	53.62°
Horizontal gap, *b*	154 mm
Pitch, *d*	1137 mm

TABLE 7

Max installed capacity (MW$_p$) and annual energy output (GWh/year).

LGAs constituting the city of Ibadan	Max. installable capacity (MW$_p$)	Annual energy output (GWh/year)
Akinyele	136.16	523.63
Egbeda	125.79	483.74
Ibadan-North	293.71	1129.5
Ibadan North-East	267.84	1030.0
Ibadan North-West	143.73	552.73
Ibadan South-East	219.42	843.82
Ibadan South-West	220.62	848.42
Ido	52.07	200.25
Lagelu	66.95	257.48
Oluyole	88.94	342.04
Ona-Ara	119.55	459.77
Total (City of Ibadan)	1734.8	6671.4

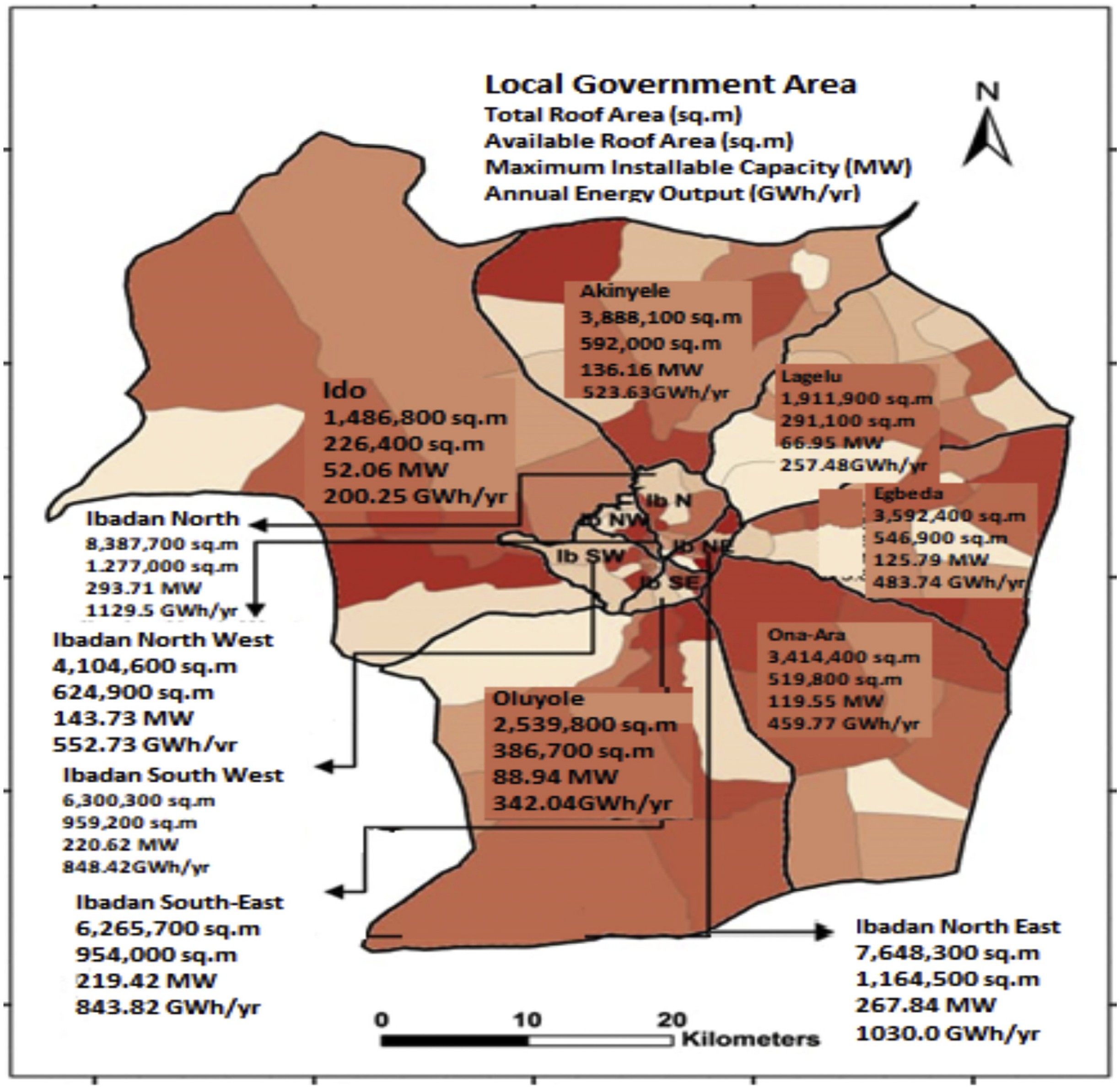

FIGURE 5

Thematic map of total roof area, available roof area, maximum installable capacity and annual energy generation potential for each of the local government areas constituting the city of Ibadan.

5.5. Annual energy output obtainable from rooftop PV system for the city of Ibadan

The maximum installed capacity as well as the annual energy output obtainable from each of the local government areas (municipalities) constituting the city of Ibadan were determined using Eqs. (30) and (31–34), respectively and the results are presented in Table 7. From the table, it can be deduced that the maximum installable capacity for the city of Ibadan is calculated to be about 1.735 GW_p, which would be able to produce an annual energy output of 6.671 TW h. According to the study by Ayodele et al. [54] in which the energy consumption of the residential buildings in the City of Ibadan was estimated to be 0.76 TW h. It can be deduced that rooftop PV has huge potential to satisfy the residential electricity needs of the city of Ibadan.

Figure 5 presents at a glance the total roof area, available roof area, maximum installable capacity and annual energy output of each of the Local Government Areas constituting the city of Ibadan. The map could be useful to the would be-investors, engineers, policy makers, government agencies, NGOs etc. for optimal planning and development of solar rooftop PV in the city of Ibadan.

6. Conclusion

The rooftop PV electricity potential has been evaluated for the city of Ibadan. The population sampling, Google Earth imagery and ArcGIS 10.4 were utilised in determining the available roof area required for PV deployment within the city. All the rooftops within the study area were assessed, first based on orientation, then shading and other uses. The inter-row panel shading was obtained as to minimize efficiency loss when one panel creates shadows behind the other. The study reveals that the optimum fixed tilt angle for rooftop PV deployment in the city of Ibadan is about 11°. The total roof area of the city was estimated as 49,540,000 m^2 (i.e. 49.54 km^2) while the available roof area for rooftop PV deployment was evaluated as 7,542,500 m^2 (i.e. 7.543 km^2) representing about 0.15 ultilization factor. The annual solar energy that could be harvested on rooftop for the city of Ibadan was estimated to be about 6.6714 TW h/year, with a maximum plant capacity of 1734.8 bvcMW$_p$. This represents annual specific energy yield of 3845.6 kW h/kW$_p$ indicating that the city of Ibadan has a fairly good solar radiation. Future research would look into economic viability as well as greenhouse gas saving potential of rooftop PV deployment for the city of Ibadan. The authors would also look into the impact of solar rooftop PV deployment on the small signal stability when integrated into the city grid network.

Credit author statement

Ayodele and Ogunjuyigbe: Conceptualization, Methodology, Software. Ayodele and **Nwakanma**: Data curation, Writing- Original draft preparation. Ayodele and **Nwakanma**: Visualization, Investigation. Supervision.: Ayodele and Ogunjuyigbe: Software, Validation.: Ayodele, **Nwakanma and** Ogunjuyigbe: Writing- Reviewing and Editing,

Conflict of interest

The authors hereby declare that there is no any conflict of interest regarding the publication of this manuscript.

References

[1] T.R. Ayodele, O.A. Olanrewaju, A.A. Jimoh, J. Sustain. Dev. Africa 14 (no. 6) (2012) 84–97.
[2] C.O. Okoye, O. Taylan, D.K. Baker, Renew. Sustain. Energy Rev. 55 (2016) 550–566.
[3] J.L. Sawin, et al., Renewables 2017 Global Status Report, 2013.
[4] A.H. Dehwah, M. Asif, M.T. Rahman, Energy Build. 171 (2018) 76–87.
[5] M. Campbell, P. Aschenbrenner, J. Blunden, E. Smeloff, S. Wright, The Drivers of the Levelized Cost of Electricity for Utility-scale Photovoltaics, White Paper: SunPower Corporation, 2008.
[6] Y. Li, D. Ding, C. Liu, C. Wang, Energy Build. 129 (2016) 563–573.
[7] N. Martín-Chivelet, Energy 94 (2016) 233–242.
[8] P. Romero-Lankao, D. Dodman, Curr. Opin. Environ. Sustain. 3 (3) (2011) 113–120.
[9] M.H. Chung, Sustain. Cities Soc. 41 (2018) 341–348.
[10] J. Melius, R. Margolis, S. Ong, Estimating Rooftop Suitability for PV: a Review of Methods, Patents, and Validation Techniques, National Renewable Energy Laboratory (NREL), Golden, CO, 2013.
[11] J. Byrne, L. Kurdgelashvili, S. Hegedus, Creating A Solar City: Determining The Potential Of Solar Rooftop Systems In The City Of Newark, Center for Energy and Environmental Policy, University of Delaware, 2009.
[12] J. Byrne, J. Taminiau, L. Kurdgelashvili, K.N. Kim, Renew. Sustain. Energy Rev. 41 (2015) 830–844.
[13] J. Hofierka, Kanuk, Renew. Energy 34 (10) (2009) 2206–2214.
[14] J. Ordonez, E. Jadraque, J. Alegre, G. Martinez, Renew. Sustain. Energy Rev. 14 (7) (2010) 2122–2130.
[15] L. Bergamasco, P. Asinari, Sol. Energy 85 (5) (2011) 1041–1055.
[16] M.C. Brito, N. Gomes, T. Santos, J.A. Tenedorio, Sol. Energy 86 (1) (2012) 283–288.
[17] R. Singh, R. Banerjee, Estimation of roof-top photovoltaic potential using satellite imagery and GIS, in: Photovoltaic Specialists Conference (PVSC), 2013 IEEE 39th, IEEE, (2013), pp. 2343–2347.
[18] N. Lukα, D. laus, S. Seme, B. alik, G. tumberger, Appl. Energy 102 (2013) 803–812.
[19] J. Yuan, C. Farnham, K. Emura, S. Lu, Urban Clim. 17 (2016) 1–19.
[20] N.M. Kumar, K. Sudhakar, M. Samykano, Energy Sources A Recovery Util. Environ. Eff. (2018) 2476–2484. , http://dx.doi.org/10.1080/15567036.2018.1502847.
[21] A.K. Shukla, K. Sudhakar, P. Baredar, R. Mamat, Sol. Energy 170 (2018) 1162–1170.
[22] L.K. Wiginton, H.T. Nguyen, J.M. Pearce, Comput. Environ. Urban Syst. 34 (4) (2010) 345–357.
[23] H. Lehmann, S. Peter, Assessment of Roof & Façade Potentials for Solar Use in Europe, Institute for sustainable solutions and innovations (ISUSI): Aachen, Germany, 2003.
[24] J. Paidipati, L. Frantzis, H. Sawyer, A. Kurrasch, Rooftop Photovoltaics Market Penetration Scenarios, National Renewable Energy Laboratory (NREL), Golden, CO, 2008.
[25] P. Denholm, R. Margolis, Supply Curves for Rooftop Solar PV-generated Electricity for the United States, 2008.
[26] H. Ladner-Garcia, E. O'Neill-Carrillo, Determining realistic photovoltaic generation targets in an isolated power system, in: Power & Energy Society General Meeting, 2009. PES'09. IEEE, IEEE, (2009), pp. 1–5.
[27] R. Vardimon, Renew. Energy 36 (no. 2) (2011) 591–594.
[28] J. Khan, M.H. Arsalan, Renew. Energy 90 (2016) 188–203.
[29] O.S. Ohunakin, M.S. Adaramola, O.M. Oyewola, R.O. Fagbenle, Appl. Energy 112 (2013) 152–159.
[30] O.I. Okoro, E. Chikuni, P. Govender, Prospect of wind energy in Nigeria, Proceedings of the International Conference on Domestic Use of Energy (2007).
[31] B. Ayeni, The Metropolitan Area of Ibadan: Its Growth and Structure, Ibadan region, 1994, 72–84.
[32] A.A. Adedeji, Spatial Exploration and Analysis of Electricity Poverty: a Case Study of Ibadan, Southwestern, Nigeria, Department of Geography, 2016.
[33] M.M. Hoogwijk, On the Global and Regional Potential of Renewable Energy Sources, 2004.
[34] A. Bahrami, C.O. Okoye, U. Atikol, Renew. Energy 113 (2017) 563–579.
[35] S.A. Mousavi Maleki, H. Hizam, C. Gomes, Energies 10 (1) (2017) 134.
[36] J. Page, The calculation of monthly mean solar radiation for horizontal and inclined surfaces from sunshine records for latitudes 40N-40S, UN Conference on New Sources of Energy (1961) 378–390, vol. 4.
[37] B. Matek, B. Schmidt, The Values of Geothermal Energy: A Discussion of the Benefits Geothermal Power Provides to the Future US Power System, in: Geothermal Energy Association, Washington DC and Geothermal Resources Council, Davis, CA, 2013, Available at: http://geo-energy.org/reports/Values%20of%20Geothermal, 20.
[38] B.Y. Liu, R.C. Jordan, Sol. Energy 4 (3) (1960) 1–19.
[39] H. Taubenböck, A. Roth, S. Dech, Linking Structural Urban Characteristics Derived From High Resolution Satellite Data to Population Distribution, Urban and regional data management, 200735–45.
[40] S. Izquierdo, M. Rodrigues, N. Fueyo, Sol. Energy 82 (10) (2008) 929–939.
[41] R. Naroll, Am. Antiq. 27 (4) (1962) 587–589.
[42] C. Pratt, Water Sci. Technol. 39 (no. 5) (1999) 145–151.
[43] National-Bureau-of-Statistics, Demographic Statistics Bulleting, 2020, pp. 1–2. (Accessed 4 April, 2021) https://www.citypopulation.de/php/nigeria-admin.php?adm1id=NGA031.
[44] M. Luqman, S.R. Ahmad, S. Khan, U. Ahmad, A. Raza, F. Akmal, Smart Grid Renew. Energy 6 (05) (2015) 128.
[45] Y. Mijinyawa, S.O. Adesogan, O.G. Ogunkoya, J. Build. Apprais. 3 (no. 1) (2007) 52–58.
[46] I.R. Pillai, R. Banerjee, Sol. Energy 81 (no. 2) (2007) 162–172.
[47] J. Schallenberg-Rodriguez, Renew. Sustain. Energy Rev. 20 (2013) 219–239.
[48] T. Toikka, Effects of Non-tracking Solar Collector Orientation on Energy Production: Photovoltaic Systems, 2012.
[49] SolarWorld, SW 280 MONO BLACK (33mm frame), 2021, pp. 1–2. (Accessed 4 April 2021) https://www.solarelectricsupply.com/solarworld-sunmodule-plus-sw-280-mono-black-solar-panel-280-watt.
[50] J.A. Duffie, W.A. Beckman, Solar Engineering of Thermal Processes, John Wiley & Sons, 2013.
[51] R. Singh, R. Banerjee, Energy Procedia 90 (2016) 401–411.
[52] E. Kjellsson, Potential for Building Integrated Photovoltaics–a Study for Sweden, IEA-PVPS Task, 20002–4.
[53] E. Ghisi, Build. Environ. 41 (no. 11) (2006) 1544–1550.
[54] T. Ayodele, A. Ogunjuyigbe, I. Atiba, Sustain. Cities Soc. 30 (2017) 171–183.